

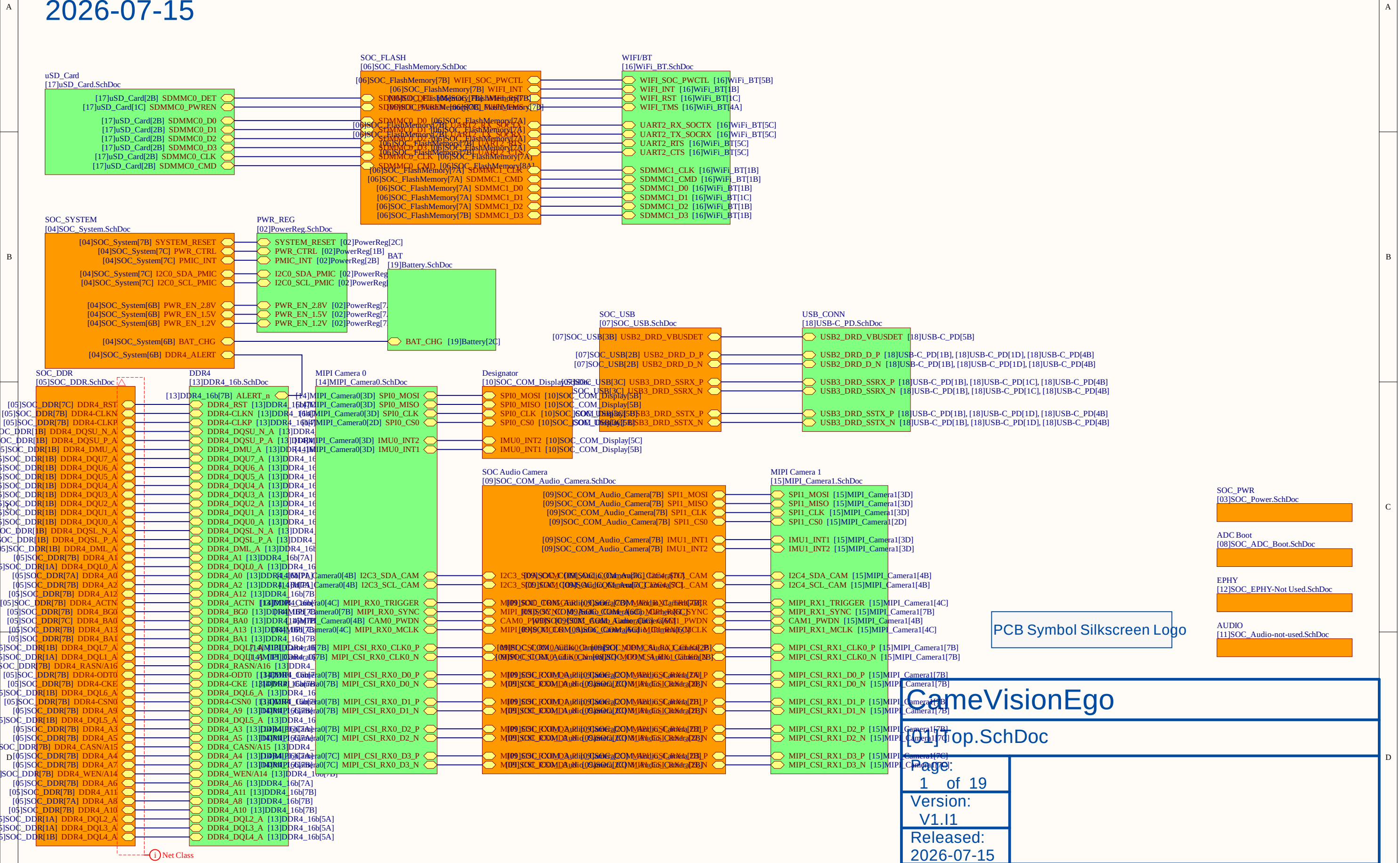
CameVisionEgo

V1.I1

2026-07-15

SOC

Peripheral



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[01]Top.SchDoc

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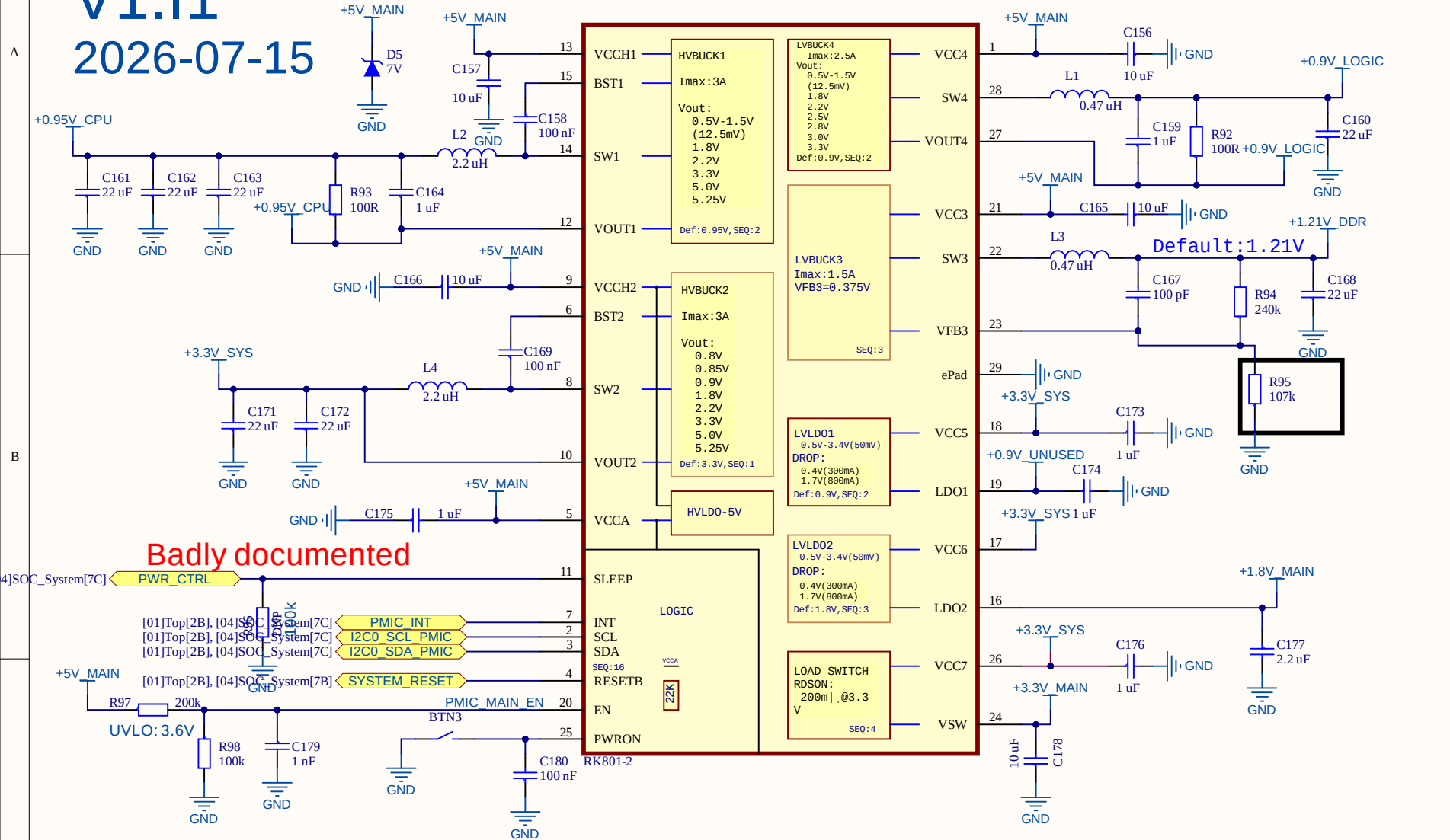
Version: V1.I1

Released: 2026-07-15

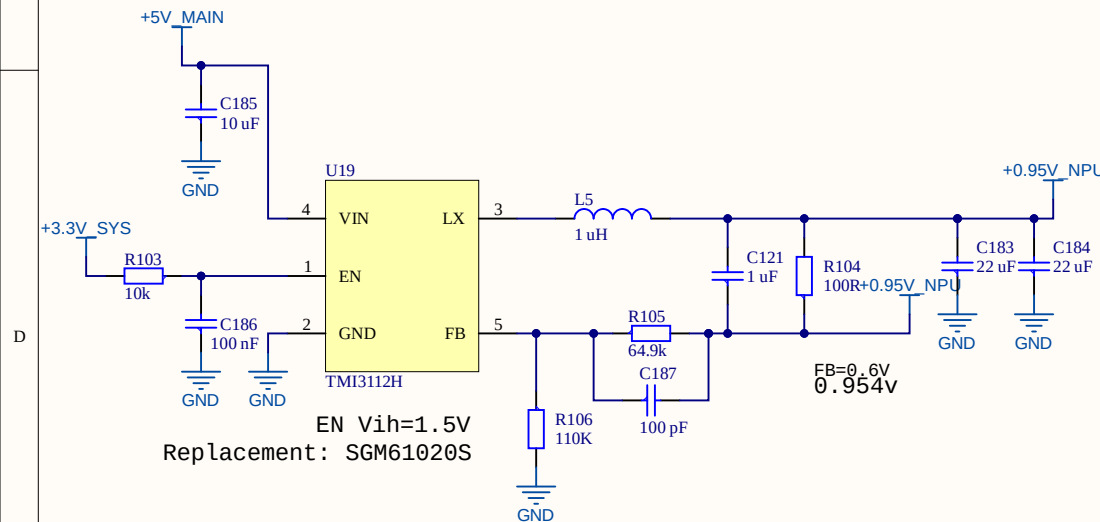
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V1.I1

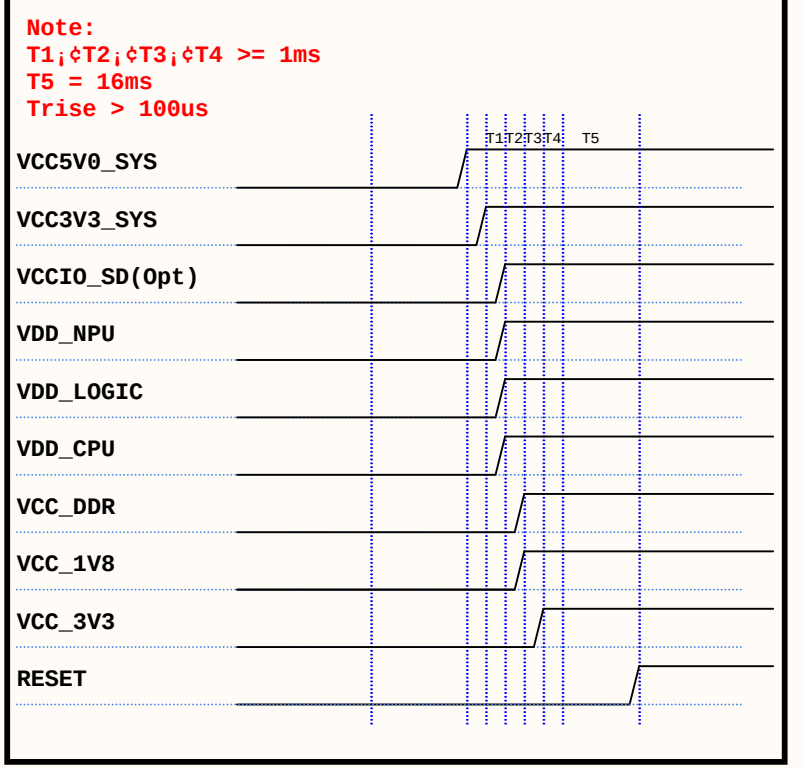
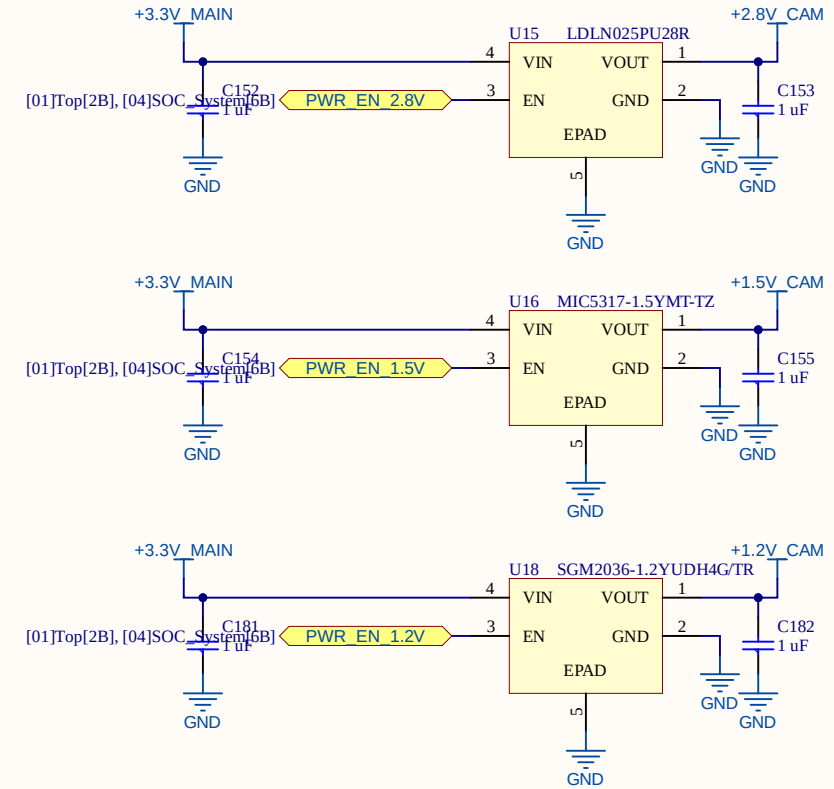
2026-07-15



Badly documented



Power Name	PMIC Channel	Time Slot	Default voltage	Peak Current
VCC5V0_SYS	EXT BUCK	Slot: 0	5.25V	
VCC3V3_SYS	RK801 HVBUCK2	Slot: 1	3.3V	
VCC10_SD(0pt)	RK801 LD01	Slot: 2	0.9V	
VCC_NPU	RK801 HVBUCK1	Slot: 2	0.95V	3A
VDD_LOGIC	RK801 LVBUCK4	Slot: 2	0.9V	1.5A
VDD_CPU	EXT BUCK	Slot: 2a	0.95V	2A
VCC_DDR	RK801 LVBUCK3	Slot: 3	1.35V	
VCC_1V8	RK801 LD02	Slot: 3	1.8V	
VCC_3V3	RK801 SWITCH	Slot: 4	3.3V	
RESET	nPOR RESET about 16ms after NPOR_DET is ready			



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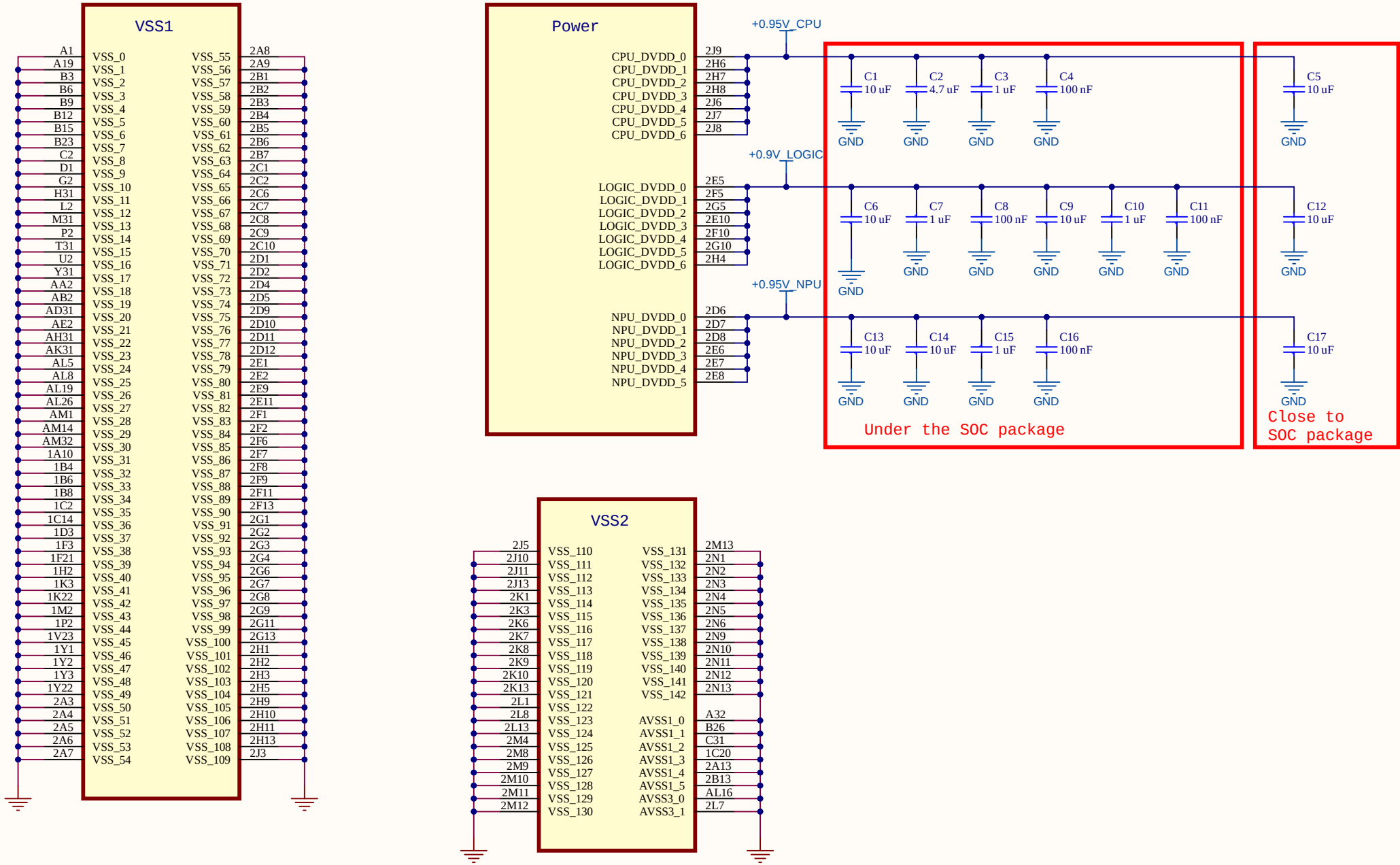
[02]PowerReg.SchDoc

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Version: V1.I1

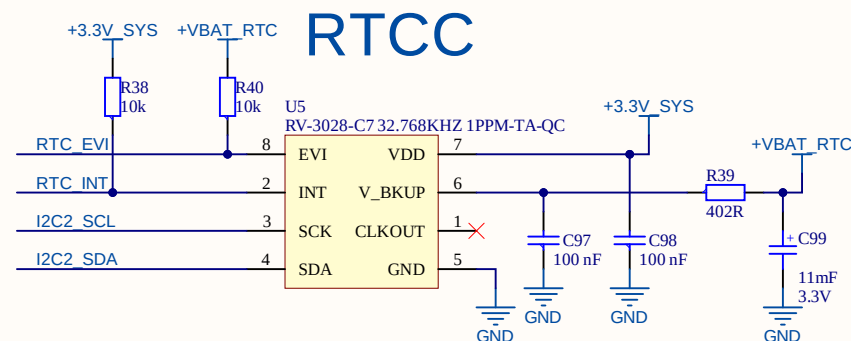
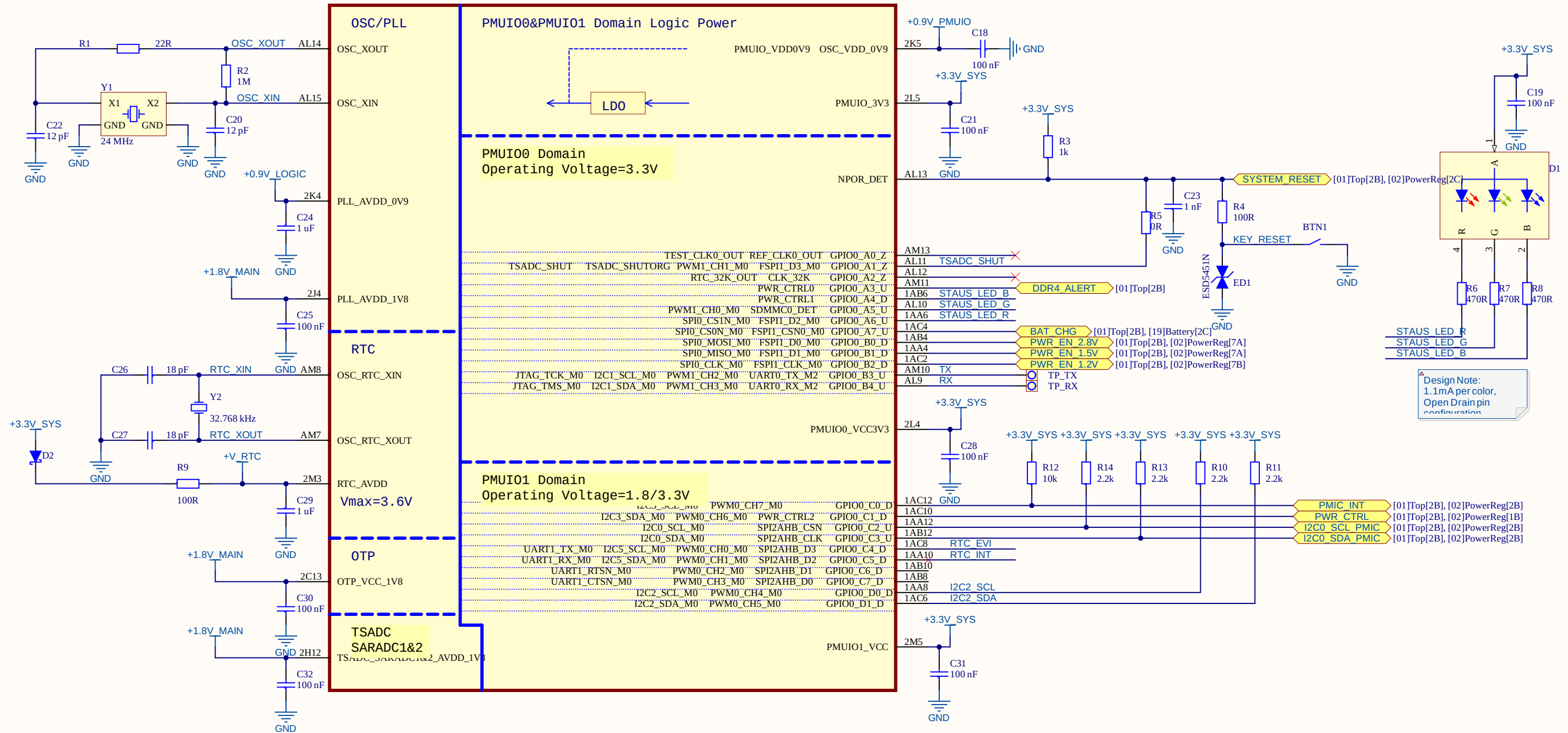
Released: 2026-07-15

SOC: Power and Ground



2026-07-15

SOC: OSC/PLL/RTC/OTP/PMU



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[04]SOC_System.SchDoc

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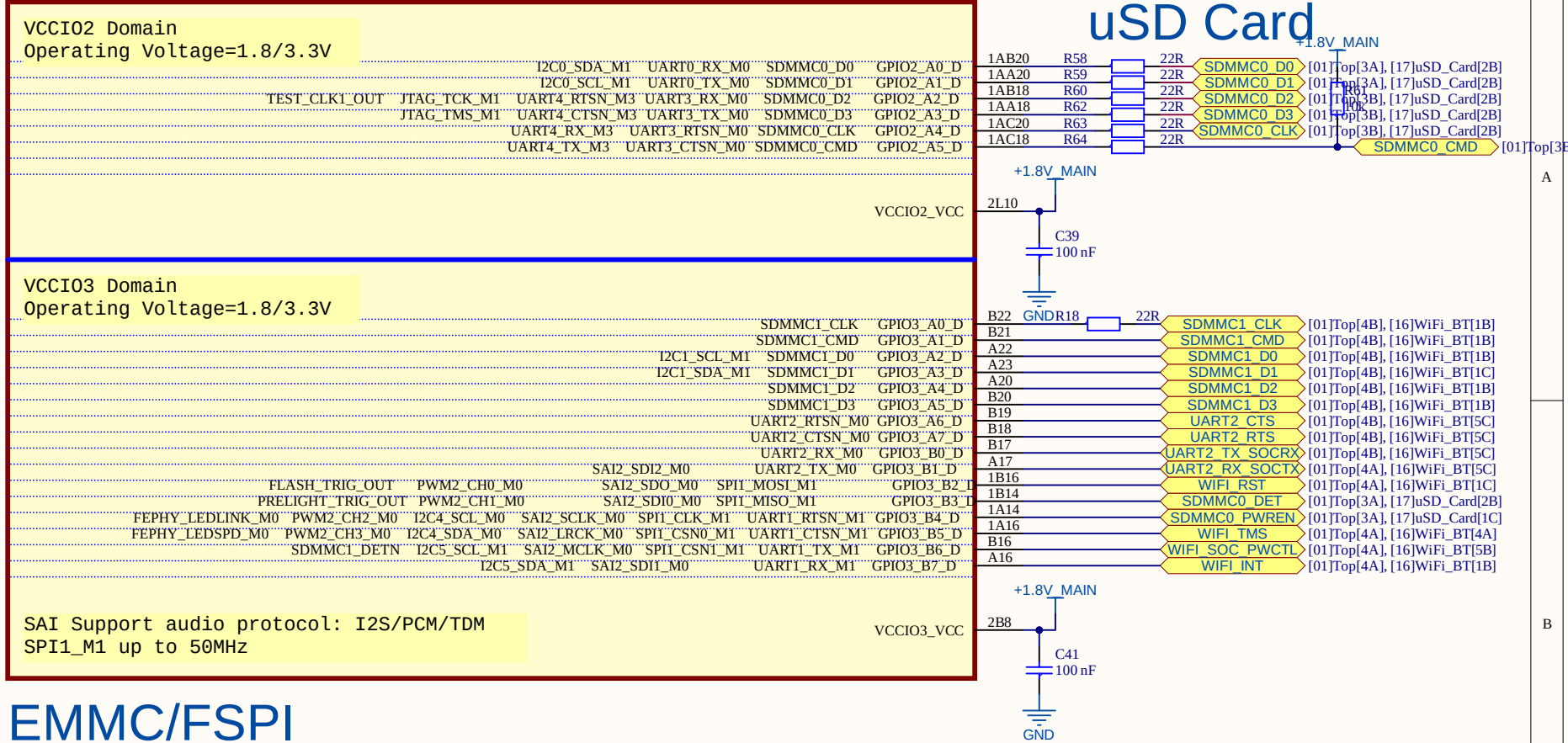
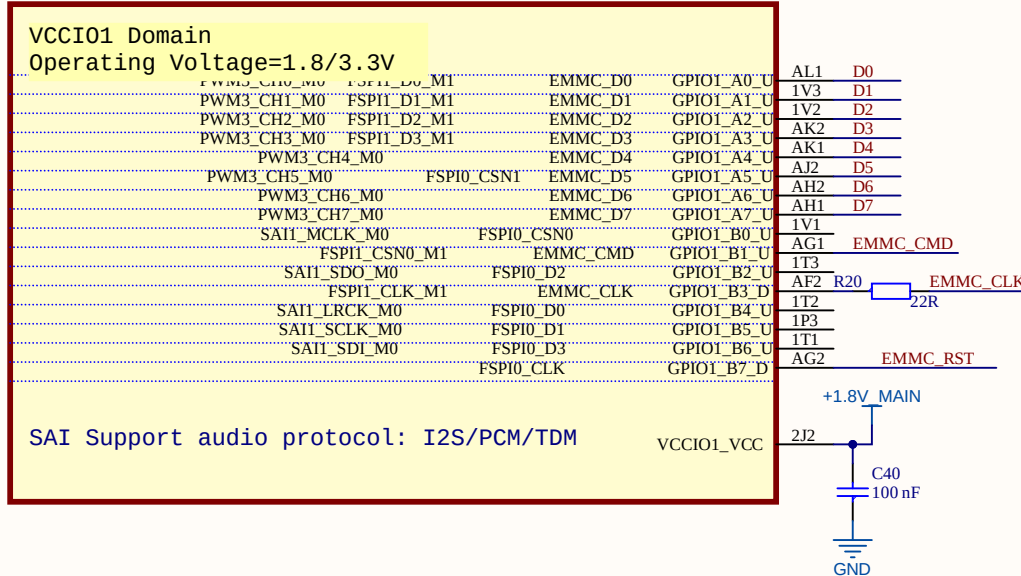
Version:
V1.11

Released:
2026-07-15

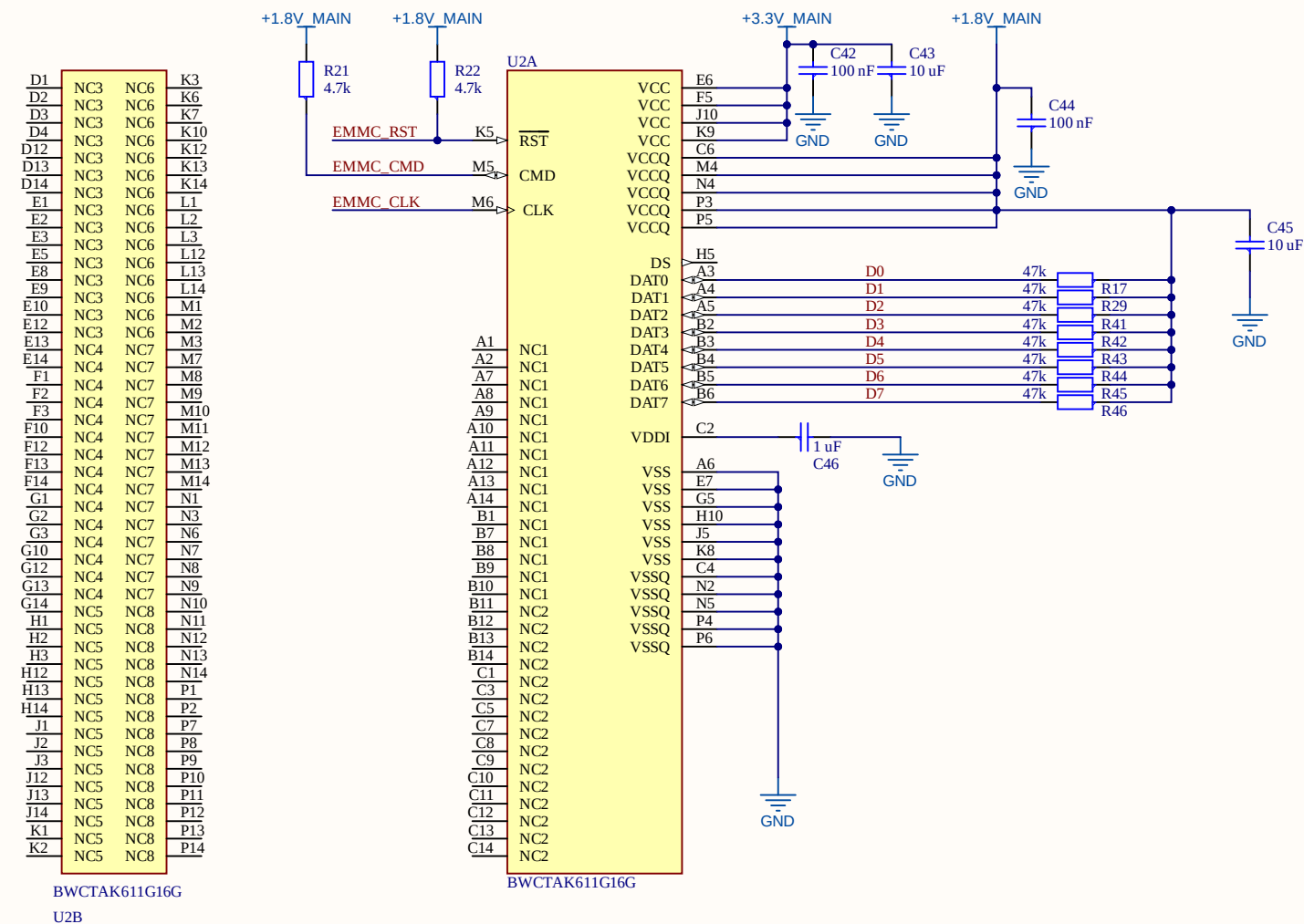
SOC: DDR Controller



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V1.11 SOC: EMMC/FSPI
2026-07-15



EMMC/FSPI



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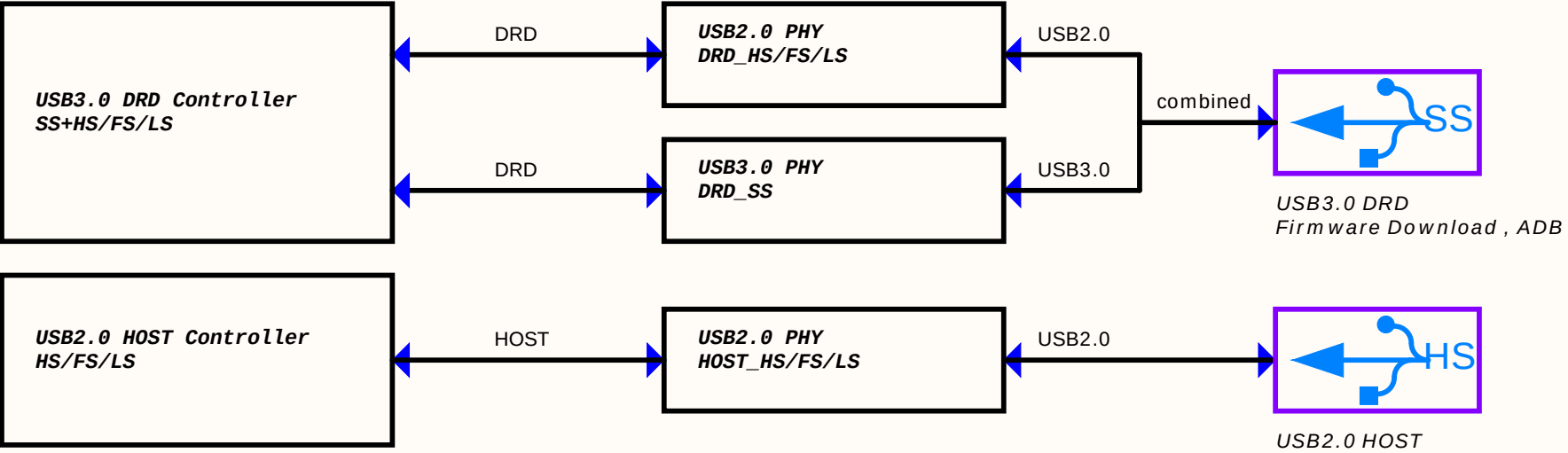
[06]SOC_FlashMemory.SchDoc

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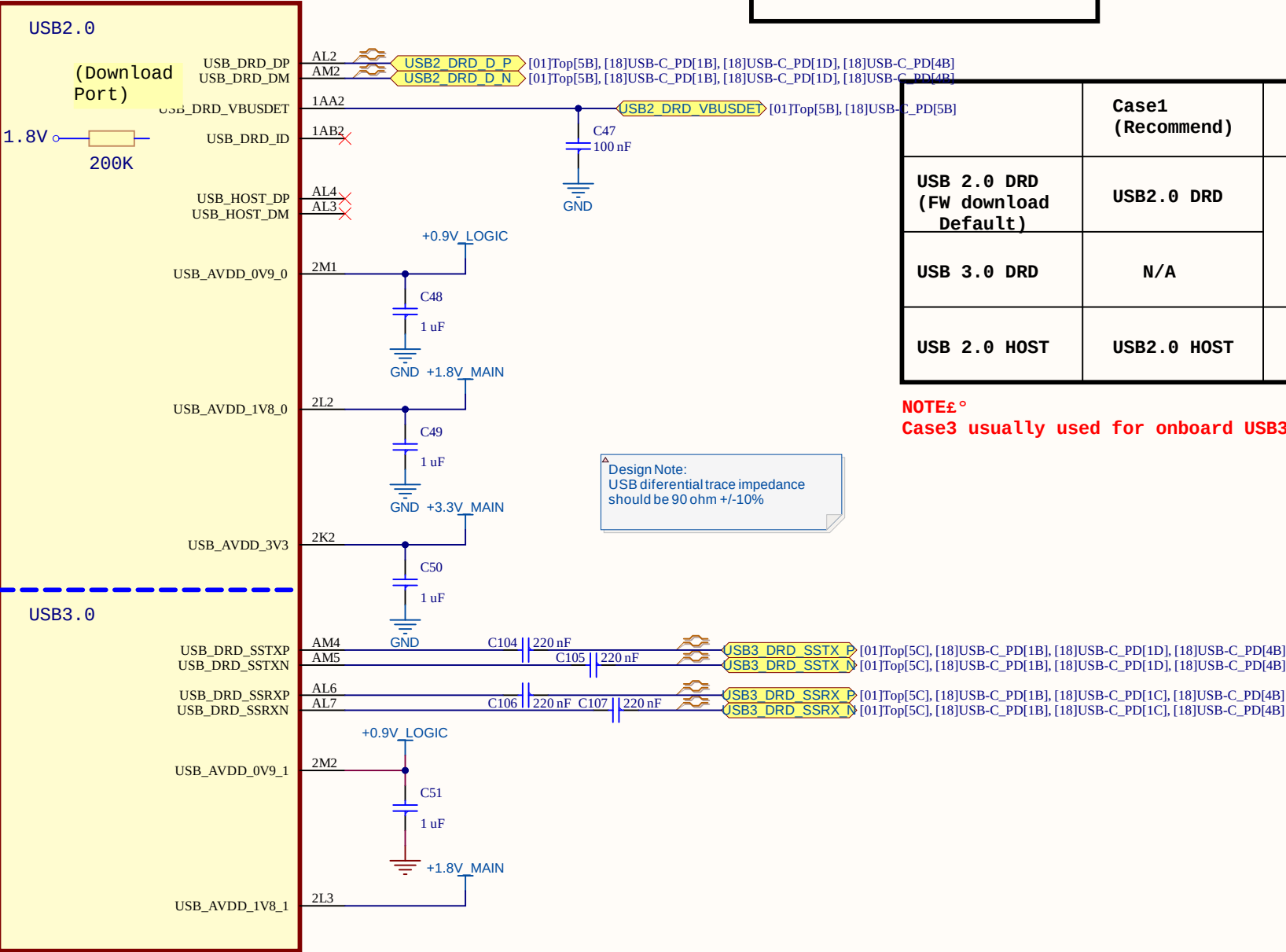
Version:
V1.11

Released:
2026-07-15

USB Map



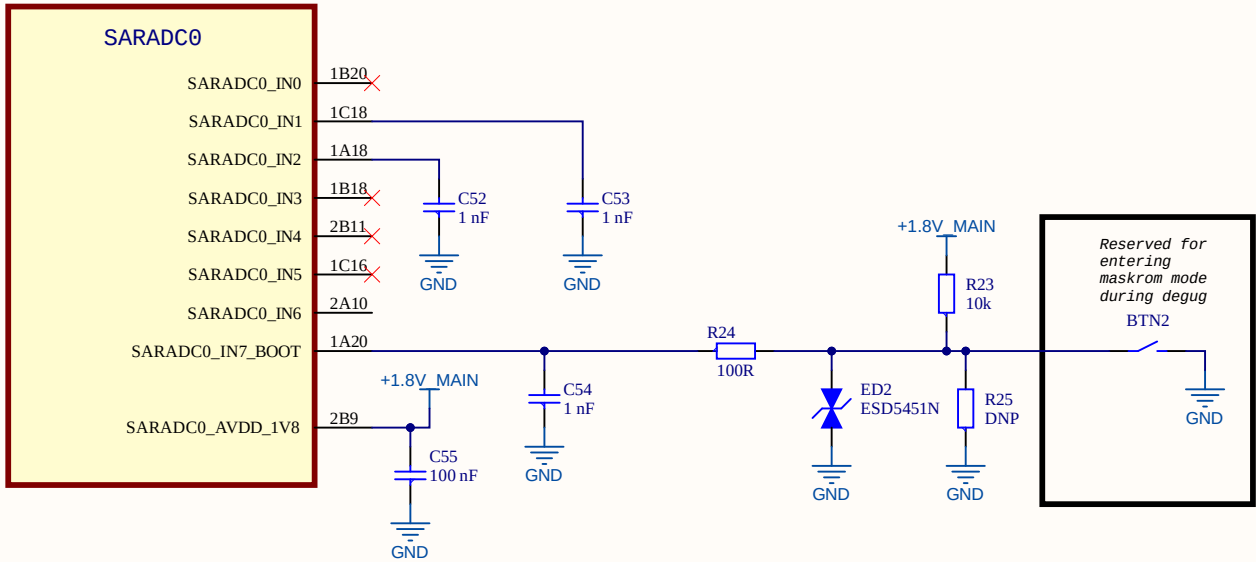
SOC: USB Controller



	Case1 (Recommend)	Case2 (Recommend)	Case3 (Not recommend)
USB 2.0 DRD (FW download Default)	USB2.0 DRD	Combined for USB3.0 DRD	USB2.0 HOST
USB 3.0 DRD	N/A		Combined for USB3.0 HOST
USB 2.0 HOST	USB2.0 HOST	USB2.0 HOST	

NOTE: Case3 usually used for onboard USB3.0 Host.

SOC: SARADC0 / BOOT



BOOT Mode Config of SARADC0_IN7

Item	Rup	Rdwon	Boot Mode
Config1	NC	10K	USB/UART0(Maskrom mode)
Config2	100K	12K/27K	FSPI0--SDMMC0--USB/UART0
Config3	100K	51K/82K	FSPI0--EMMC--USB/UART0
Config4	100K	120K	EMMC--USB/UART0
Config5	100K	200K	EMMC--SDMMC0--USB/UART0
Config6	100K	330K/820K	SDMMC0--USB/UART0
Config7	10K	NC	FSPI0--EMMC--SDMMC0--USB/UART0(Auto mode)

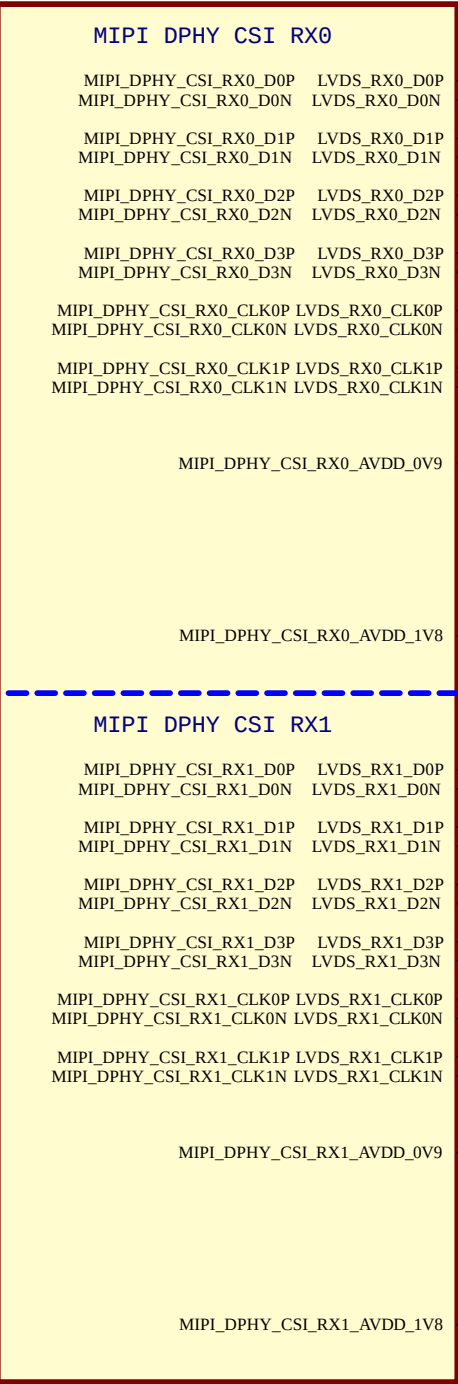
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V1.I1

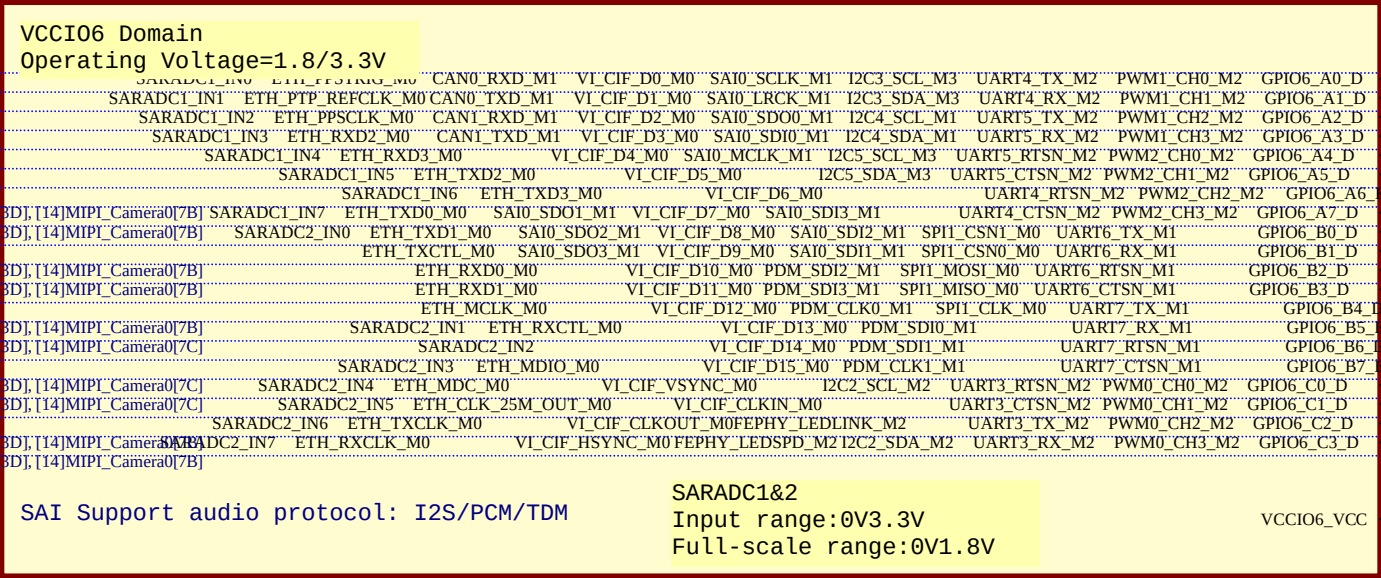
2026-07-15

SOC: VI_CIF/ETH/SAI/UART/PWM/I2C

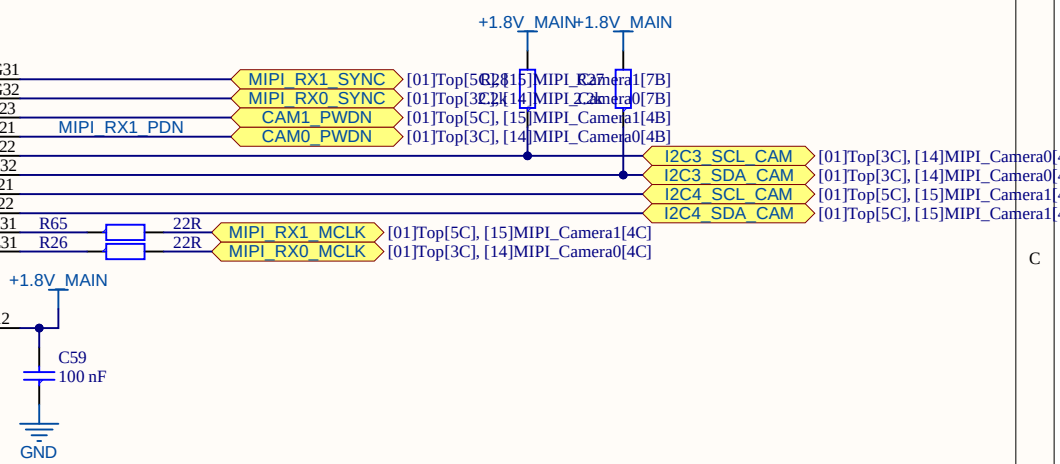
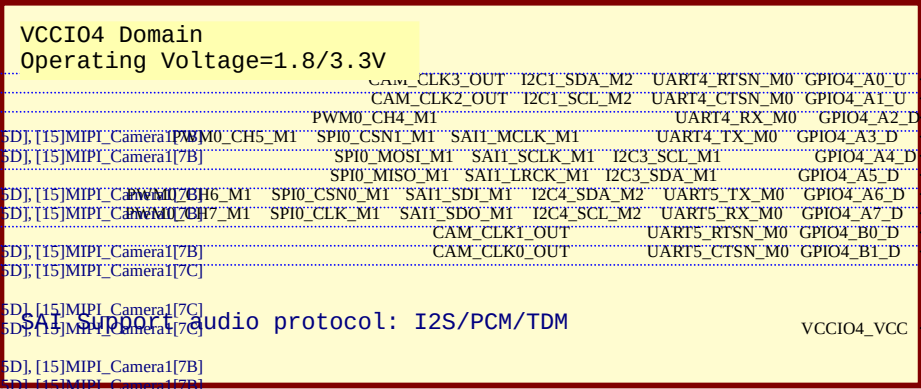
SOC: MIPI-CSI



Design Note:
MIPI diferential trace
impedance should be 100 ohm



SOC: I2C/SPI/MIPI-CLK



Note:

MIPI CSI RX design rules:

- Trace impedance 100ohm+/-10%
- The distance between other signals follows the 3W rule.

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[09]SOC_COM_Audio_Camera.SchDoc

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Version:
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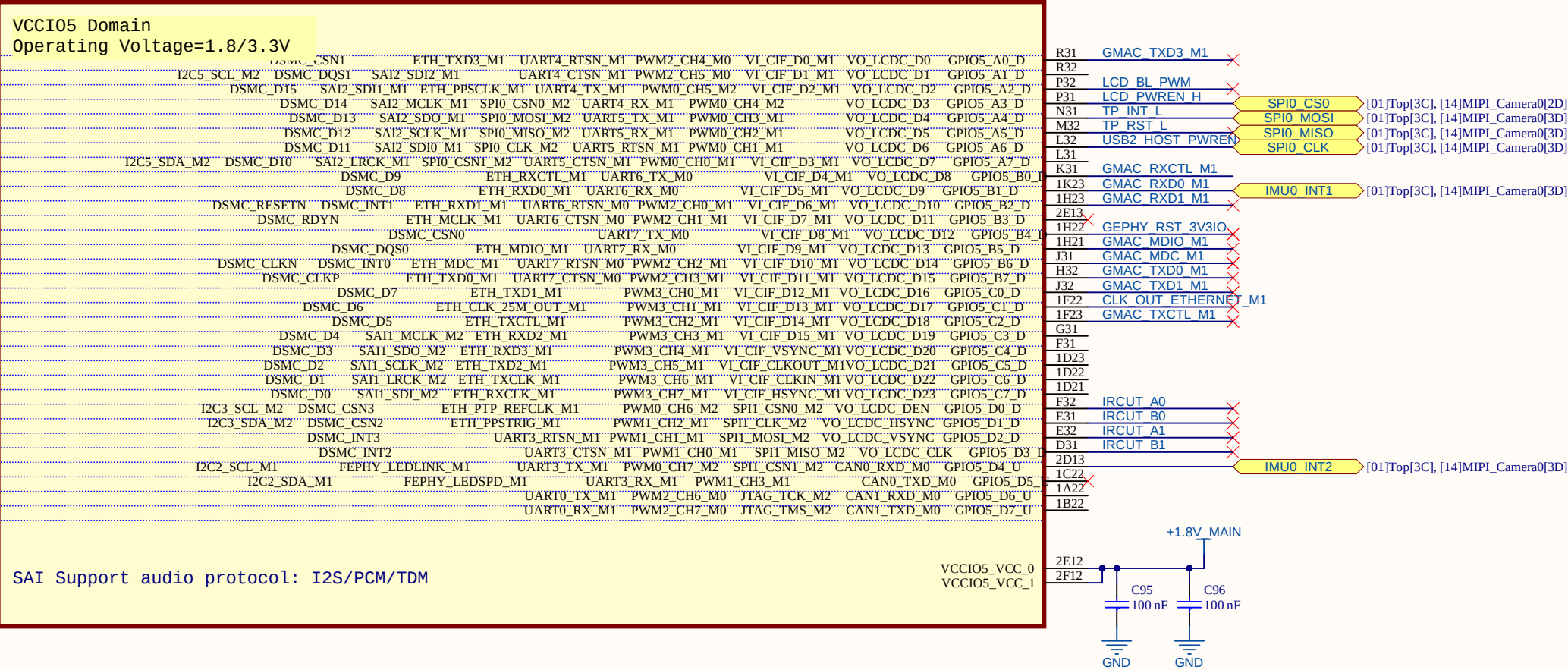
Released:
2026-07-15

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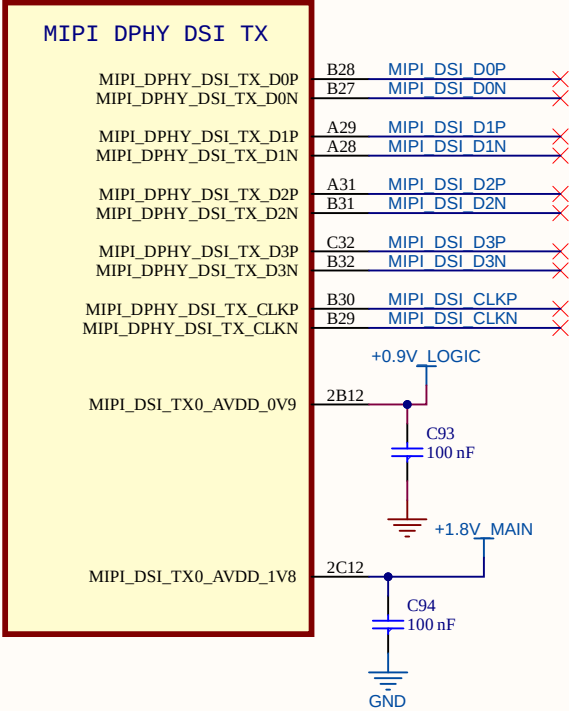
V1.I1

2026-07-15

SOC: VO_LCDC/VI_CIF/ETH/SAI/UART/PWM/I2C



SOC: MIPI-DSI Interface



Note:

MIPI DSI RX design rules:

- Trace impedance 100ohm+/-10%
- The distance between other signals follows the 3W rule.

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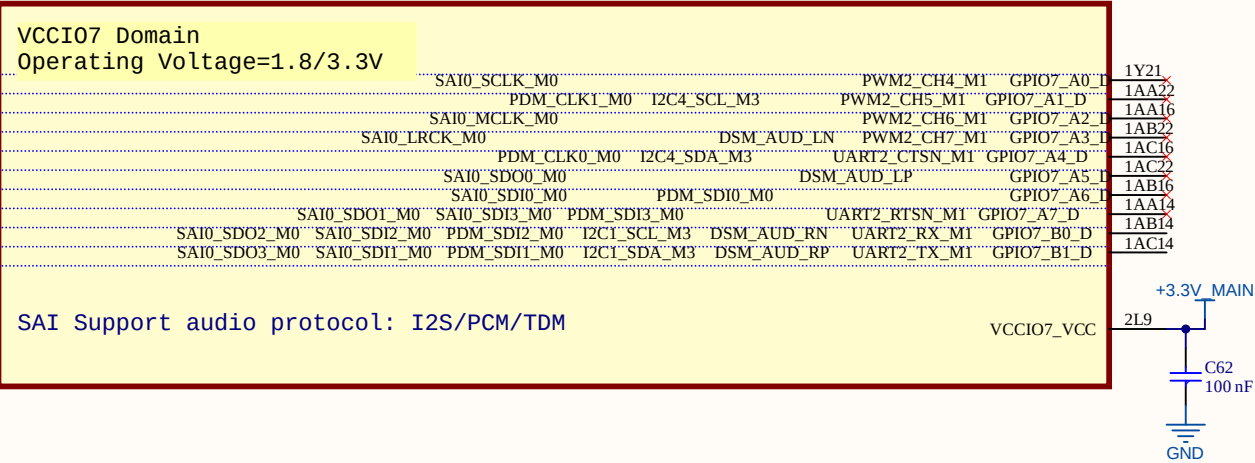
[10]SOC_COM_Display.SchDoc

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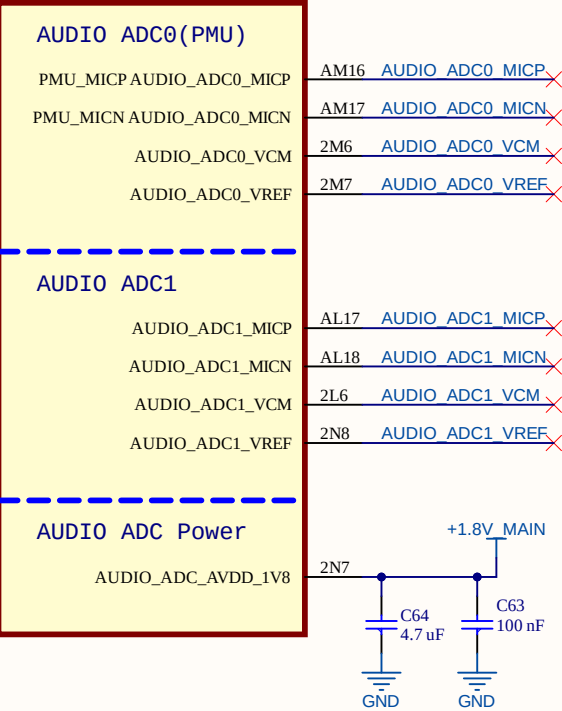
Version:
V1.I1

Released:
2026-07-15

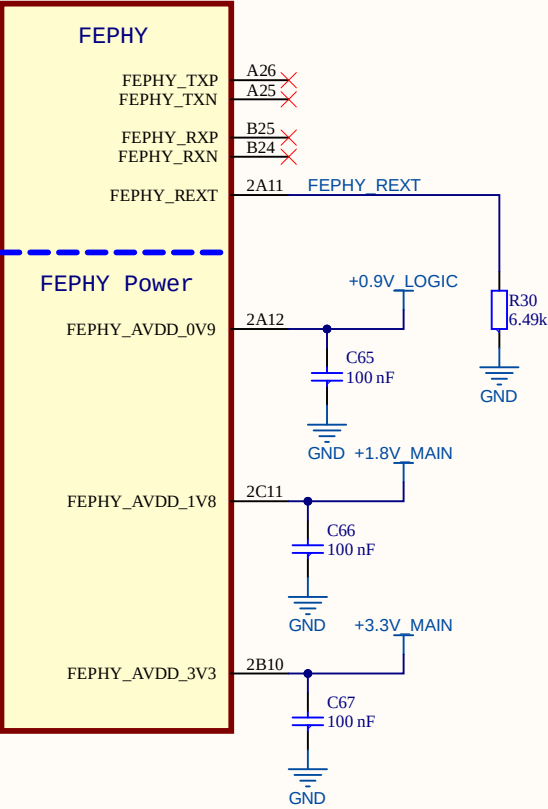
SOC: Audio Interface



SOC: Audio ADC



SOC: FEPHY

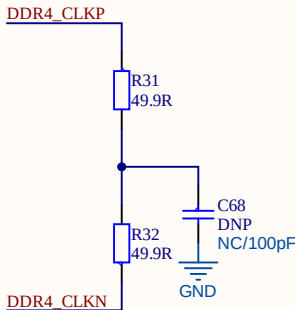


Note:

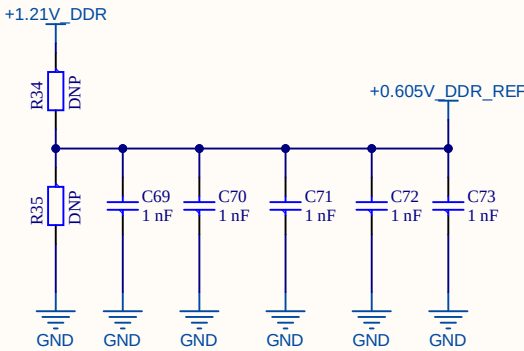
MIPI DSI RX design rules:

- Trace impedance 100ohm+/-10%
- The distance between other signals follows the 3W rule.

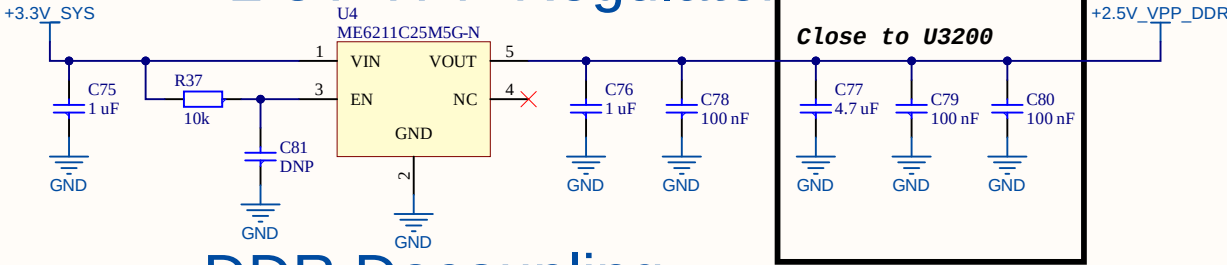
Signal Conditioning



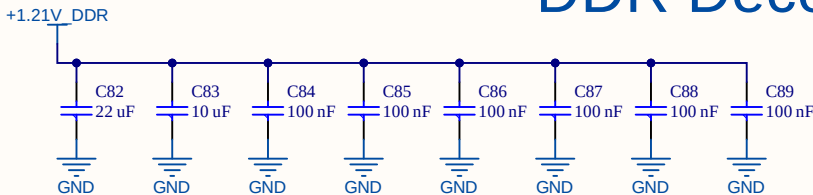
VREF stabilization



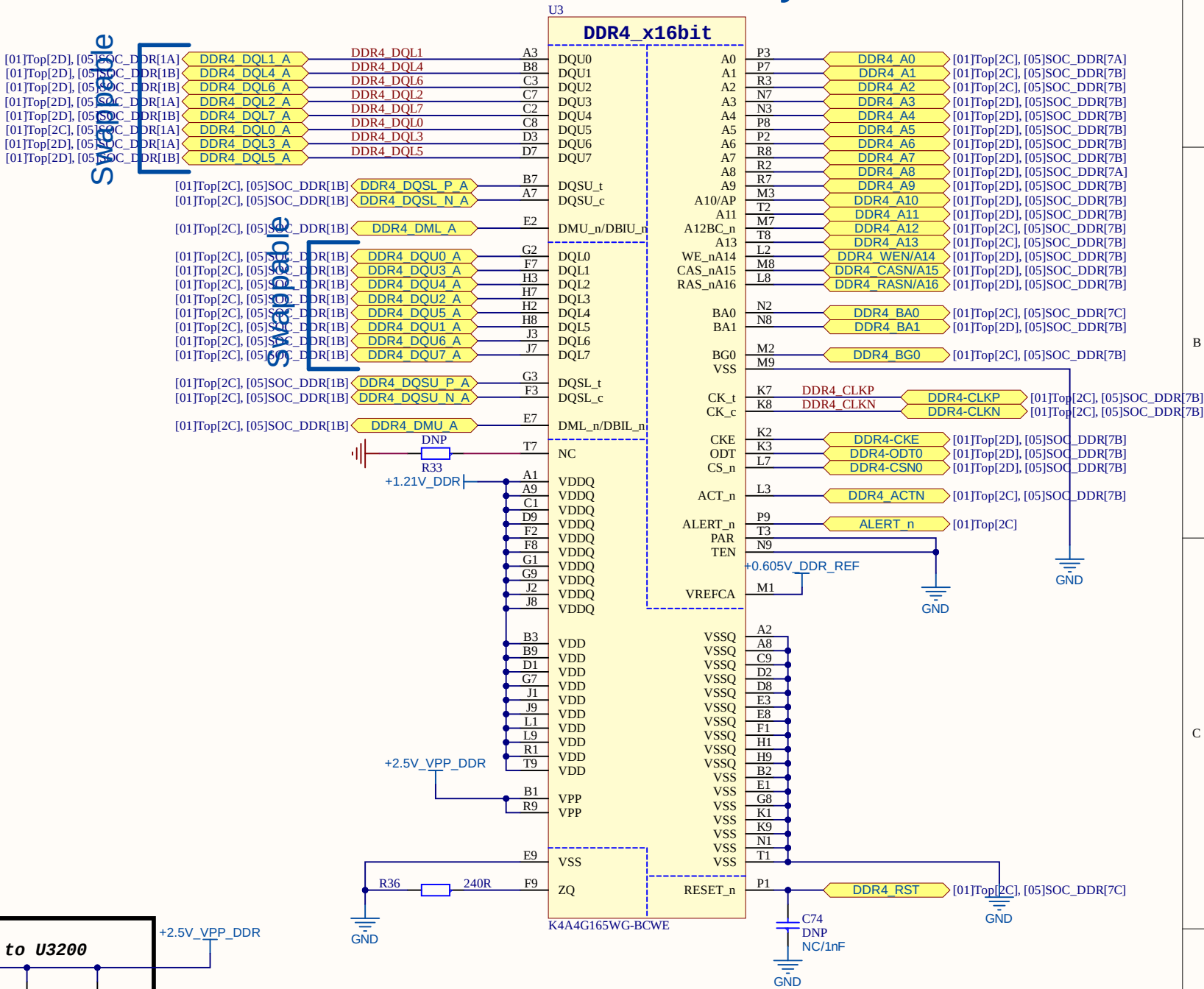
+2.5V VPP Regulator



DDR Decoupling



DDR Memory



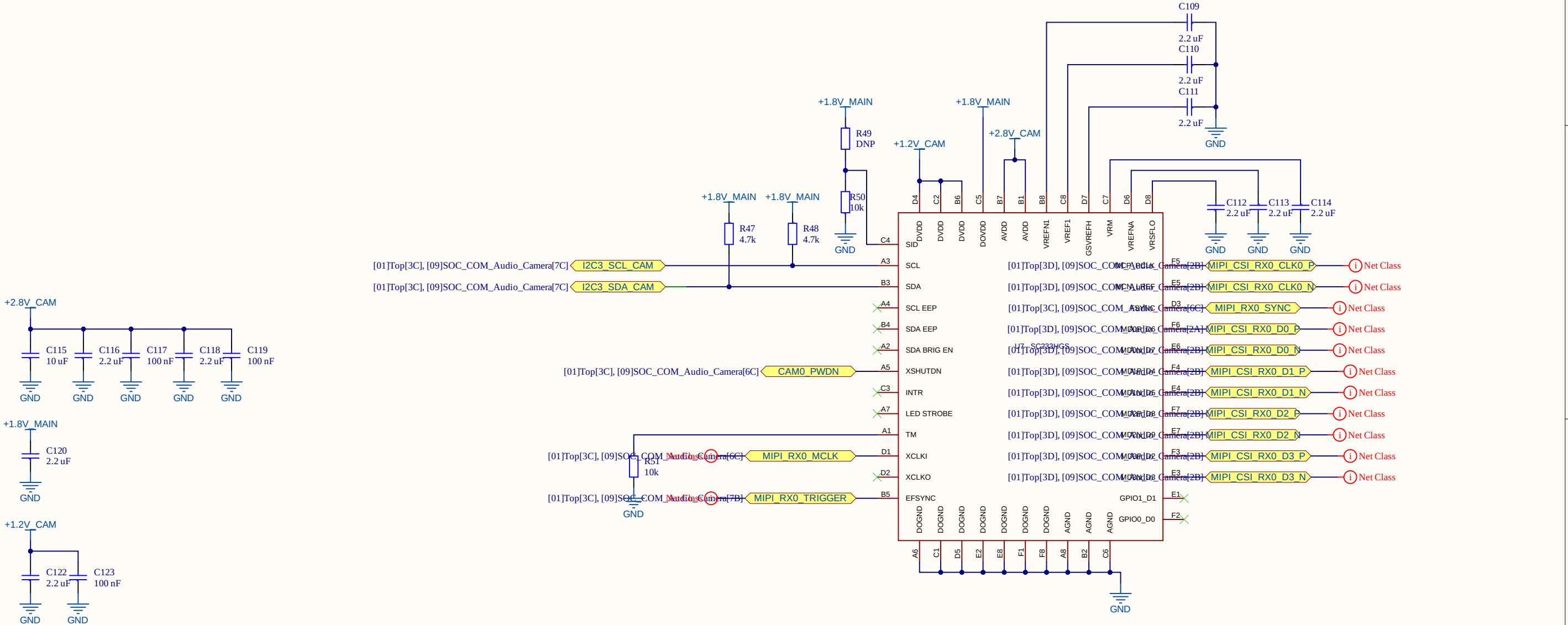
CameVisionEgo

V1.I1

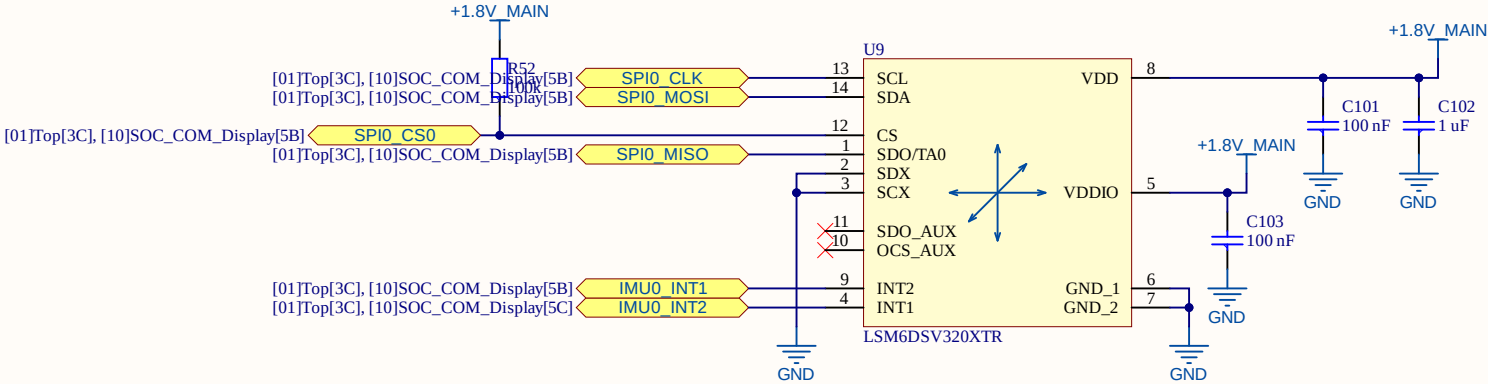
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MIPI Camera Interface

Cam 0



Cam 0 IMU



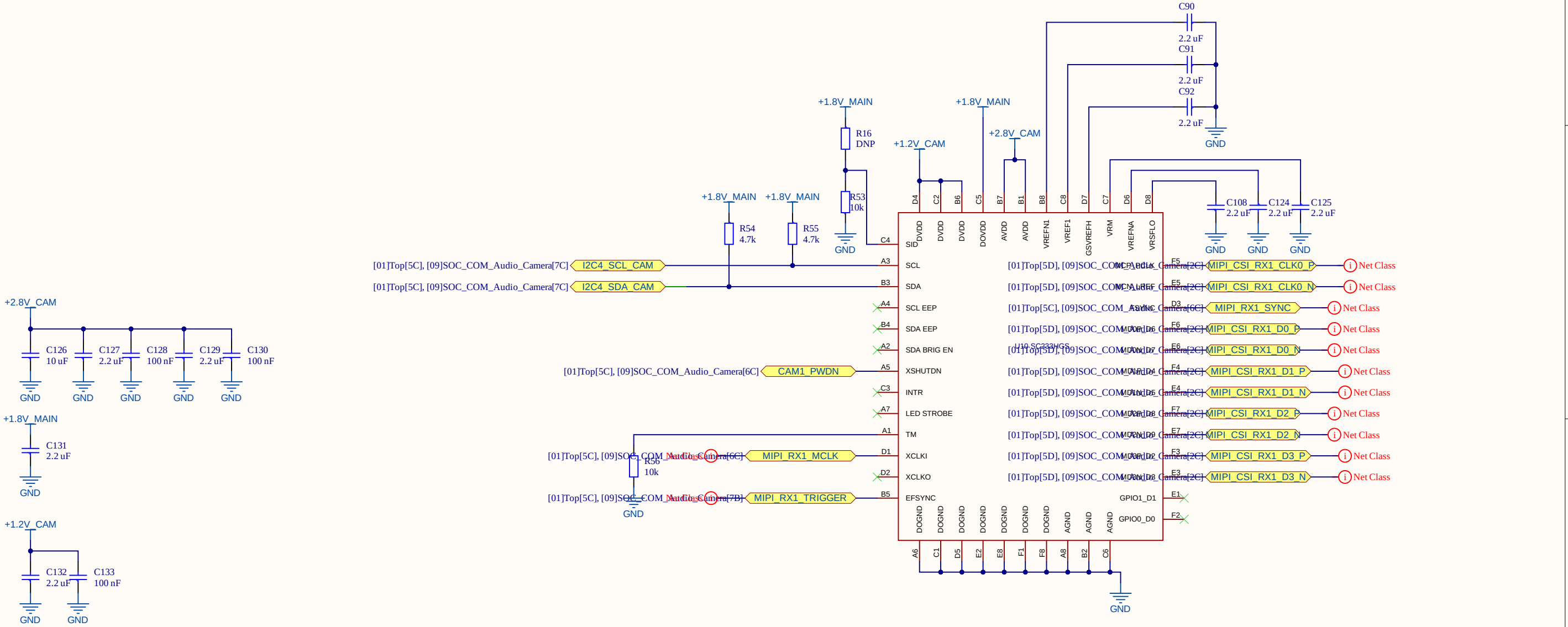
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V1.I1

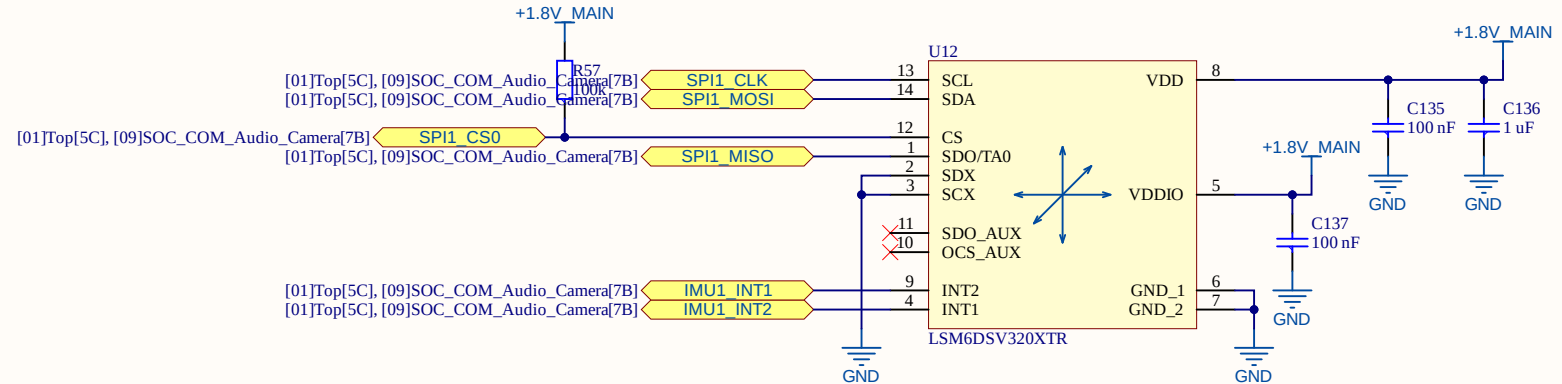
2026-07-15

MIPI Camera Interface

Cam 1



Cam 1 IMU



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[15]MIPI_Camera1.SchDoc

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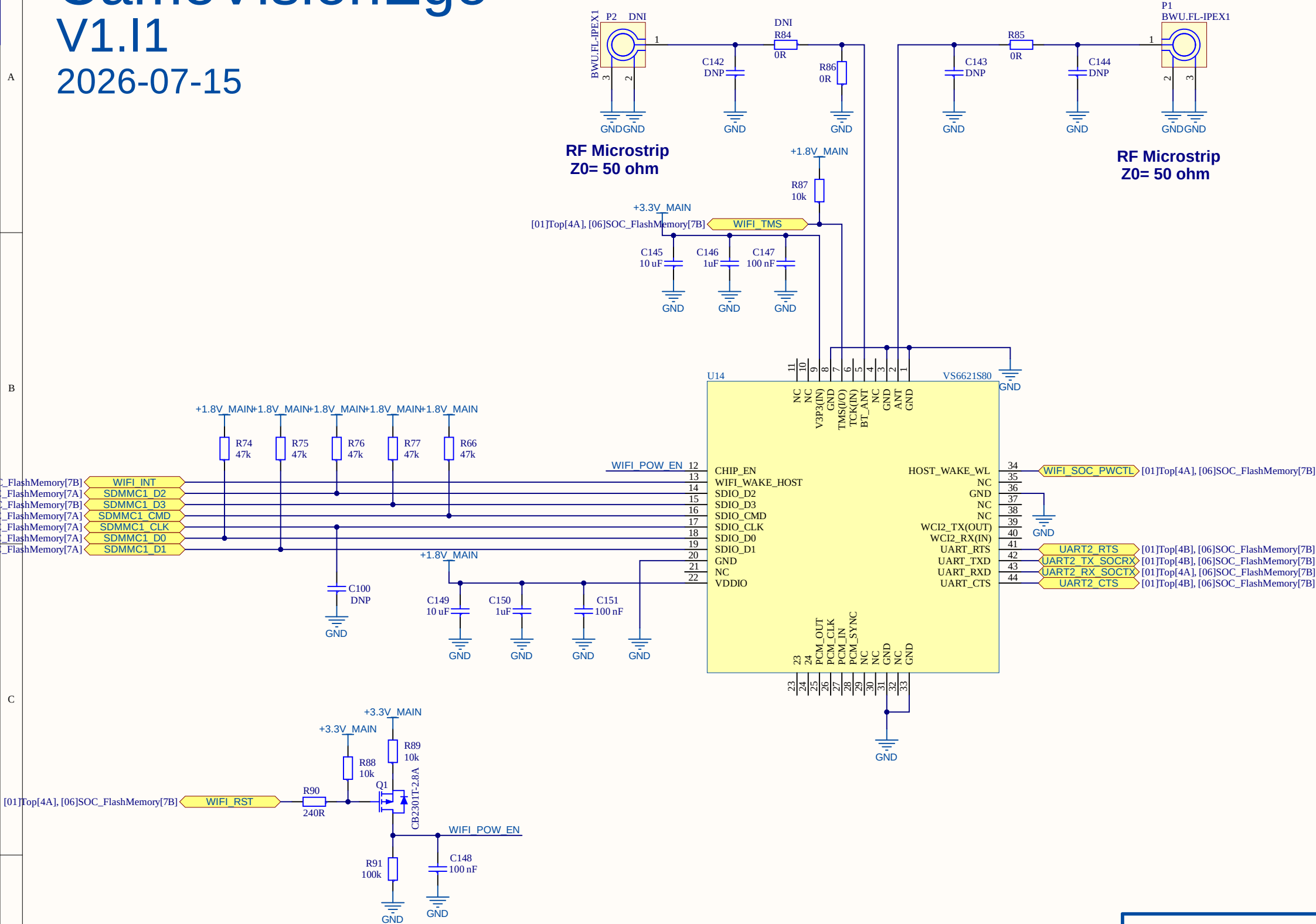
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V1.I1

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WiFi & BT Module



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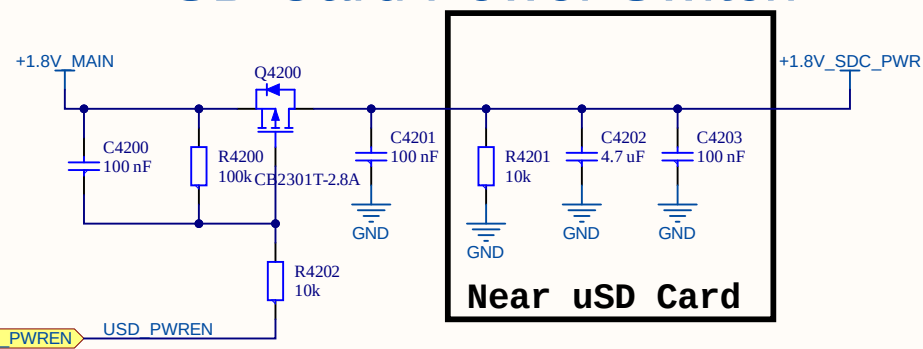
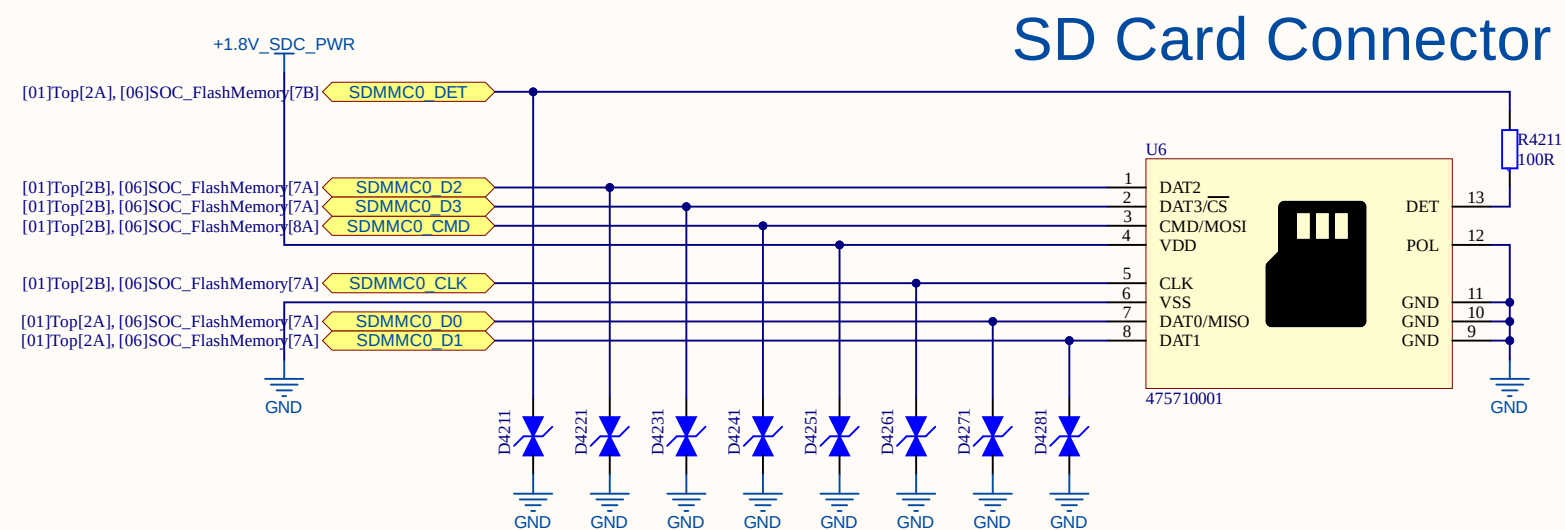
[16]WiFi_BT.SchDoc

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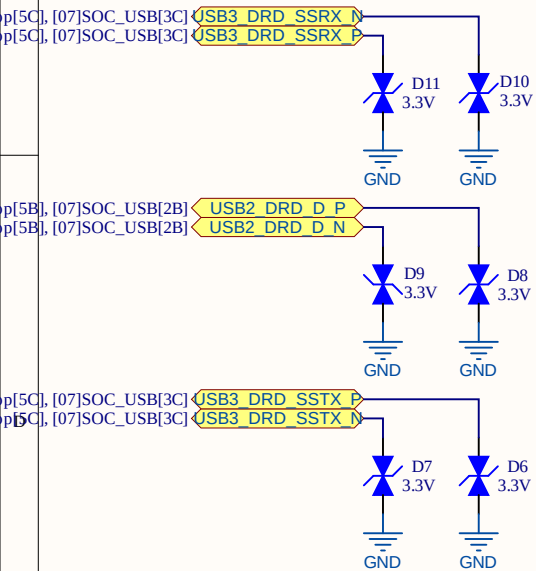
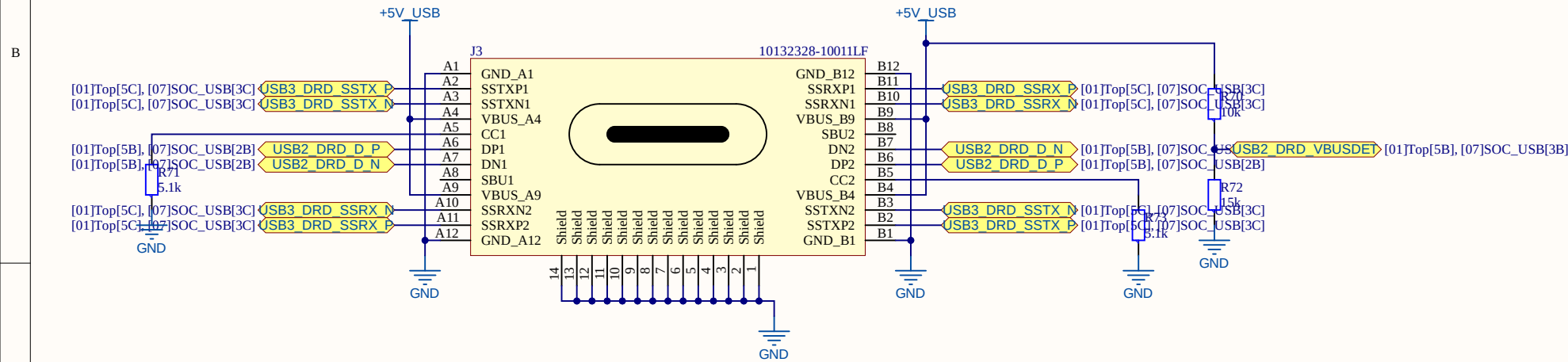
Released:
2026-07-15

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V1.11
2026-07-15



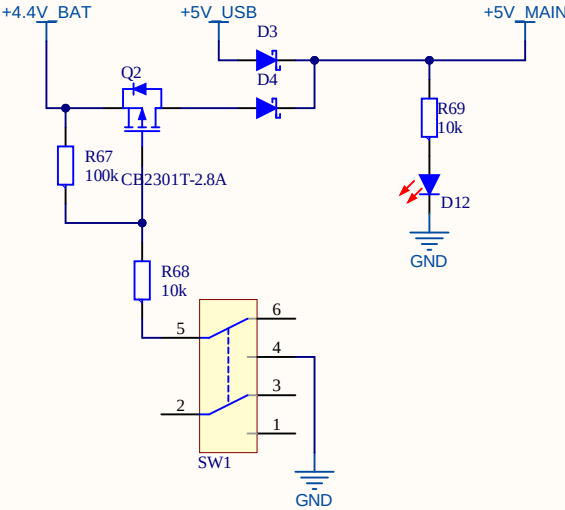
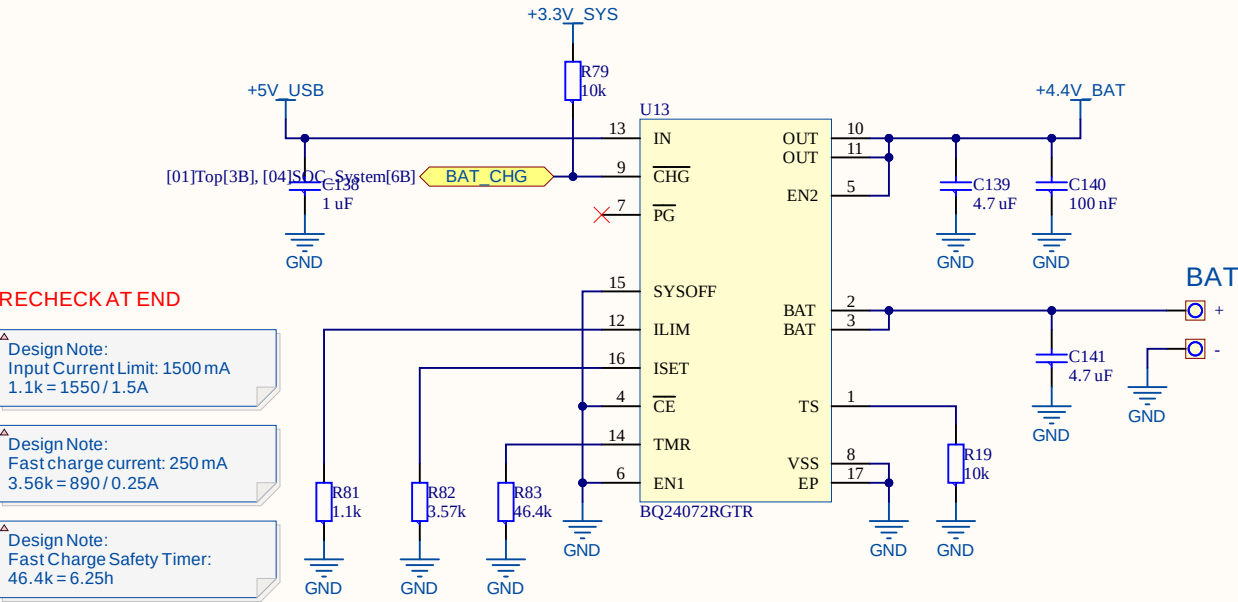
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[17]uSD_Card.SchDoc	
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Version: V1.11	
Released: 2026-07-15	

USB-C Connector



Battery Charger

Power Path



Design Note:
Power path is integrated into battery charger, this is backup plan to use the board without it